Notes on Analysis

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1 Definition of Real Number and Least Upper Bound

Remark. We need to add

1. Order of $\mathbb{Q}$.
2. archimedean property.
3. Or can we prove LUB with density of $\mathbb{Q}$ and if we have monotone+bounded $\implies$ Cauchy for rational sequences.

Let $\mathbb{N}$ be natural number and $\mathbb{Z}$ be integers and $\mathbb{Q}$ be rationals, which are introduced by adding subtraction $-$ and devision $/$.

Definition 1.1 (Sequence)**.** A sequence (x_n) in a topological space $\mathbb{F}$ is defined as a mapping $\mathbb{N} \longrightarrow \mathbb{F}$.

Definition 1.2 (Cauchy Sequence in $\mathbb{Q}$)**.** Let (x_n) be a sequence in $\mathbb{Q}$, then (x_n) is **Cauchy** if for any rational $\varepsilon > 0$, there exists N such that $|x_n - x_m| < \varepsilon$ for all $n, m > N$. Let $\mathbf{C}(\mathbb{Q})$ be the set of all Cauchy sequences in $\mathbb{Q}$.

Definition 1.3 (Equivalent Cauchy Sequence)**.** If $(x_n), (y_n)$ be Cauchy sequneces in $\mathbb{Q}$, then define $\sim$: $(x_n) \sim (y_n)$ if for any rational $\varepsilon > 0$, there exists N such that $|x_n - y_n| < \varepsilon$ for all $n > N$.

Definition 1.4 (Real Numbers)**.** Define $\mathbb{R} = \mathbf{C}(\mathbb{Q})/\sim$.

Theorem 1.5 (Least Upper Bound Property of $\mathbb{R}$)**.** *If nonempty $E \subset \mathbb{R}$ and there exists an upper bound $M \in \mathbb{R}$ of E, then $\exists \sup E \in \mathbb{R}$.*

Proof. We need an order concept of $\mathbb{R}$, Archimeadian property. Suppose for fixed $n \in \mathbb{N}$, there exists K such that $K/n > M$ and $L/n < x_0 \in E$, then there exists unique m_n such that m_n/n is an upper bound of E and $\frac{m_n - 1}{n}$ is not an upper bound. So we can construct two equivlent Cauchy sequences $\left(\frac{m_n}{n}\right)$ and $\left(\frac{m_n - 1}{n}\right)$. The associated real number is the $\sup E$. $\square$

2 Convergence of Real Sequence and Series

Definition 2.1 (Convergence of Real Sequence)**.** Let $(x_n) \subset \mathbb{R}$, if there exists x such that for all $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that for all $n > N$, $|x_n - x| < \varepsilon$, then we say $\lim x_n = x$ or $x_n \longrightarrow x$ as $n \longrightarrow \infty$.

Definition 2.2 (Convergence of sequences in metric spaces)**.** Let (x_n) be a sequence in a metric space (X, d), then (x_n) converges if there exists x such that

$$\lim_{n \longrightarrow \infty} d(x_n, x) = 0$$

The lim operator is closed under addition, multiplication, division(not by zero). In fact any continuous function at the limit point will preserve the lim operator.

Definition 2.3 (Sequentially Continuous functions)**.** $f : (X, d) \longrightarrow (Y, \rho)$ is continuous at x if for any $x_n \longrightarrow_d x$, $f(x_n) \longrightarrow_\rho f(x)$.

Remark. In metric space, sequential continuity is the same as continuity.

Let $(x_n) \subset \mathbb{R}$, $S_N := \sum_{n=1}^{N} x_n$, if S_N converges, then we define $\sum_{n=1}^{\infty} x_n := \lim_N S_N$.

Theorem 2.4 (Completeness and Absolute Convergence)**.** *Let x_n be a sequence in a metric space, then X is complete if for any x_n such that $\sum |x|_n$ converges implies $\sum x_n$ converges.*

Proposition 2.5 (Comparison, Ratio and Root Tests)**.**

3 Differentiation

Definition 3.1 (Limit of a Function)**.** Let $f : X \longrightarrow Y$ be a function from a metric space to another metric space. Then if there exists $y \in Y$ such that $f(x) \longrightarrow y$ when $x \longrightarrow x_0$ then we write $\lim_{x \longrightarrow x_0} f(x) = y$. This can be generalized to topological spaces.

Definition 3.2 (Continuous Functions)**.**

Definition 3.3 (Differentiation)**.** Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$, if there exists a linear operator A such that

$$\lim_{x \longrightarrow x_0} \frac{f(x) - f(x_0) - A(x)}{|x - x_0|} = 0$$

Then we say that f is *strongly* differentiable at x_0.

Definition 3.4 (Directional Derivatives)**.**

Definition 3.5 (Higher-Order Derivatives)**.**

Theorem 3.6 (Intermediate Value Theorem)**.** *Let $f : [a, b] \longrightarrow \mathbb{R}$ be continuous, then the range of f is an interval in $\mathbb{R}$.*

Proof. It's a corollary of the fact that continuous functions map connected sets to another connected set and in $\mathbb{R}$ connected sets are intervales. □

Theorem 3.7 (Mean Value Theorem and Mean Value Inequality)**.**

Theorem 3.8 (L'Hospital Theorem)**.** *Let $f, g : [a, b] \longrightarrow \mathbb{R}$ be differentiable on $[a, b]$. If $f(a) = g(a) = 0$ and $g(x)' \neq 0$ for all $x \in [a, b]$. If $\lim_{x \longrightarrow a} \frac{f'(x)}{g'(x)} = A$, then $\lim_{x \longrightarrow a} \frac{f(x)}{g(x)} = A$ as well.*

Proof. First we show that $g(x) \neq 0$ for all $x \neq a$. This is because $g'(x) \neq 0$.

Then we use Mean Value by considering $h_x(y) = f'(y)g(x) - f(x)g'(y)$, then $h_x(a) = h_x(x) = 0$, differentiable wrt y, so there exists y^* such that $h_x(y^*) = 0$. □

Theorem 3.9 (Taylor's Theorem)**.** *Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be k-th continuously differentiable at $x_0 \in \mathbb{R}^n$, then there exists $R_k : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ such that,*

$$f(x) = P_k(x) + R_k(x)|x - x_0|^k$$

where $P_k(x) = \sum_{|\alpha|<k} \partial^\alpha f(x_0)(x - x_0)^\alpha$ and $R_k(x) \longrightarrow 0$ as $x \longrightarrow x_0$.

Proof. □

4 Riemann Integral

Definition 4.1 (Piecewise Constant Functions)**.**

Definition 4.2 (Riemann Integral)**.**

Theorem 4.3 (Riemann Integrability Conditions)**.** *There are some useful facts about when functions are Riemann integrable.*

1. $f + g, cf, f * g, \max, \min$ *preserve integrability.*

2. *Uniformly continuous functions are Riemann integrable.*

3. *Monotone functions are Riemann integrable.*

Theorem 4.4 (Second Fundamental Theorems of Calculus)**.** *If $F'(x) = f(x)$ for every $x \in [a, b]$ and $f(x)$ is integrable, then $\int_a^b f' = F(b) - F(a)$.*

Proof. □

Theorem 4.5 (First Fundamental Theorem of Calculus)**.** *If f is Riemann integrable on $[a, b]$, then $F(x) := \int_a^x f$ is continuous. In addition, if f is continuous at x_0, then F is differentiable at x_0 and $F'(x_0) = f(x_0)$.*

Theorem 4.6 (Taylor's Theorem with Integral Remainder)**.**

5 Holomorphic Functions and Cauchy's Integral Theorems

6 Uniform Convergence

Definition 6.1 (Uniform Convergence)**.** Let (f_n) be a sequence of functions between metric spaces $X \longrightarrow Y$, then we say $f_n \rightrightarrows f$ on $E \subset X$, if for any $\varepsilon > 0$, there exists N such that for any $x \in E$, $n > N$,

$$|f_n(x) - f(x)| \leq \varepsilon$$

So the uniform convergence depends on both (f_n) and the domain E we consider.

Theorem 6.2 (Uniform Convergence and Limits)**.**

Theorem 6.3 (Weierstrass's M test)**.**

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